BY ORDER OF THE COMMANDER
JOINT BASE ANDREWS

JOINT BASE ANDREWS INSTRUCTION 21-105

22 MAY 2024

Maintenance

CRASHED DAMAGED/DISABLED AIRCRAFT RECOVERY (CDDAR) PROGRAM

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: This publication is available on the e- Publishing website at www.e-Publishing.af.mil for downloading or ordering.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: MXG/MXM

Certified by: 89 MXG/CC
(Col Bryan L. Bobeck)

Supersedes: JBANDREWSI 21-105, 16 April 2019

Pages: 31

This publication establishes responsibilities and procedures for the recovery of aircraft involved in a ground or air incident/accident within a 50-mile radius of Joint Base Andrews and ensures adequate coverage 24 hours a day, 7 days a week. It provides guidance in accordance with (IAW) AFPD 21-1, *Air and Space Maintenance*, DAFI 21-101, *Aircraft and Equipment Maintenance Management* and DAFI 21-101 *AMC Supplement 1¸ Aircraft and Equipment Maintenance Management*. This OI will be utilized in conjunction with other agency policies and all applicable Technical Orders (TOs) pertaining to the disabled aircraft. It applies to all Joint Base Andrews (JBA) organizations and personnel that maintain aircraft, aircraft systems, equipment, support equipment, and components regardless of AFSC.

Refer recommended changes and questions about this publication to the OPR listed above using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate chain of command. Requests for waivers must be submitted to the OPR listed above for consideration and approval. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Manual (AFMAN) 33- 363, *Management of Records*, and disposed of in accordance with Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This instruction has been significantly revised and should be reviewed in its entirety.

Chapter 1—GENERAL CDDAR PROGRAM INFORMATION 4

1.1. General. 4

1.2. Terms Defined IAW TO 00-80C-1. 5

Chapter 2—ORGANIZATIONAL RESPONSIBILITIES 6

2.1. IAW DAFI 21-101 AMC Sup 1. 6

2.2. 316 WG/CC: 6

2.3. 89 AW/CC. 6

2.4. 89 MXG/CC. 6

2.5. 89 MXG Executive Airlift Maintenance Contractor. 6

2.6. 89 MXG Aircraft Maintenance Contractor CDDAR Program Manager. 7

2.7. 89 MXG Aircraft Maintenance Contractor CDDAR Team Chief. 9

2.8. 316 WG Command Post. 9

2.9. 89 MXG Maintenance Operations Center (MOC) Senior Controller. 10

2.10. 89 OSS/OSAA (Airfield Management). 10

2.11. 89 AW/SE. 11

2.12. 89 AW/XP. 11

2.13. 89 AW/IGI. 11

2.14. 89 APS: 11

2.15. 316 MDG/CC. 11

2.16. 316 MDG Bioenvironmental Flight (BE). 12

2.17. 316 CES. 12

2.18. 316 CONS. 13

2.19. 316 LRS. 13

2.20. 316 FSS. 13

2.21. 316 CPTS. 13

2.22. 316 SFG. 14

2.23. 316 WG/JA. 14

2.24. JBA Units with Flying Missions. 14

Chapter 3—CDDAR TEAM QUALIFICATIONS AND SPECIALTY REQUIREMENTS 16

3.1. General. 16

3.2. CDDAR Qualifications. 16

3.3. Training Requirements. 16

3.4. Specialty Requirements. 16

Chapter 4—CDDAR VEHICLE AND SUPPORT EQUIPMENT REQUIREMENTS 18

4.1. General. 18

4.2. Minimum Vehicle/SE requirements. 18

Chapter 5—JBA DISASTER RESPONSE FORCE STRUCTURE 19

5.1. General. 19

5.2. Crisis Action Team (CAT). 19

Chapter 6—CRASH/MISHAP RESPONSE CHRONOLOGICAL PROCEDURES 20

6.1. First Responders. 20

6.2. Emergency Responders. 20

6.3. Specialized Teams. 20

Chapter 7—CDDAR AIRCRAFT RECOVERY/RECLAMATION/REMOVAL 22

7.1. Procedures. 22

Chapter 8—RESPONSE TO COMPOSITE MATERIALS 24

8.1. Initial Response to Composite Materials. 24

8.2. Composite Material Handling. 24

8.3. Containment of Composite Materials. 25

Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 26

Attachment 2—CDDAR AIRCRAFT RECOVERY CHECKLIST (POC: TEAM CHIEF) 28

Attachment 3—SAFETY BRIEF CHECKLIST 31

Chapter 1

GENERAL CDDAR PROGRAM INFORMATION

1.1. General.

1.1.1. AFI 10-2501, TO 00-80C-1, TO 00-20-1, DAFI 21-101 and all supplements are the overarching governing directives. Tenant units will be responsible to their HHQ DAFI 21-101 Supplement to ensure CDDAR compliance. This publication provides supplemental guidance for aircraft recovery/removal in the event of crash/disabled aircraft at JBA. The CDDAR program is designed to recover a crashed, damaged, or disabled aircraft in a minimum amount of time, with consideration given to the requirements of opening runways for operational use, preventing secondary damage to the aircraft, and preserving evidence for mishap or accident investigation. Refer to Joint Base Andrews IEMP 10-2 for related response requirements. The CDDAR program applies to all host and tenant organizations on Joint Base Andrews (JBA) with the following considerations:

1.1.1.1. The requirement to reopen the runway for operational use.

1.1.1.2. Prevention of unnecessary secondary damage.

1.1.1.3. Preservation of evidence for accident and safety investigation.

1.1.1.4. Safety of personnel involved with initial response and recovery operations.

1.1.1.4.1. WARNING: Incidents involving all transient aircraft with hazardous materials and/or a composite structure may cause serious injury or death to those in contact with it. The mishap aircraft owning unit will be contacted to determine composite material risks and requirements for Personal Protective Equipment (PPE). The Bioenvironmental Engineering Flight (316 OMRS/SGXB 240-857-3380), located at Joint Base Andrews, will also validate PPE requirements, and determine health hazards/risks on-site. For emergencies during off-duty hours, contact the Command Post for the on-call Bioenvironmental Engineering (BE) technician or contact technician directly at (240) 327-0415.

1.1.1.5. Only required vehicles and personnel will respond to aircraft emergencies. Responding vehicle operators must be certified to drive on the JBA airfield and possess a valid JBA airfield driver's license. Vehicles will position themselves behind responding crash vehicles and not impede movement or vision. Personnel not required to respond will clear the area not to interfere with emergency operations. If necessary, the Senior Fire Official (SFO) or Incident Commander (IC) will direct the Security Forces to clear the area of non-essential personnel. In addition, only authorized agencies approved by the JBA Fire Chief may possess, operate, and communicate radio traffic using the fire net.

1.1.1.6. During initial response to an aircraft emergency, the SFO is the IC and will determine if the aircraft is secure and safe prior to releasing the aircraft for maintenance, recovery, or safety investigation (if required). IAW Technical Order (TO) 00-80C-1, the IC has overall authority and responsibility for conducting incident operations and is responsible for the management of all incident operations at the incident site. Until the aircraft has been released by the IC, no personnel will approach the aircraft without direction from the IC. All vehicles, except Fire Emergency Services vehicles, will remain

clear of the aircraft. This does not prohibit essential vehicles (such as tow vehicles) from positioning themselves nearby for immediate use. The IC must release the aircraft or direct specific actions before any vehicles, other than a Fire Emergency Services vehicle, approach the aircraft. This restriction does not prevent emergency actions prior to Fire Emergency Services arrival.

1.1.1.7. All response vehicles (except Fire Emergency Services) and nonessential vehicles will remain clear of emergency aircraft until Fire Emergency Services and crash recovery actions are complete.

1.1.1.8. Rapid Removal of Aircraft on a Runway or Taxiway. During normal flying periods, disabled aircraft will be removed as quickly and safely as possible after touchdown.

1.1.1.9. Damaged aircraft will be removed as soon as possible depending on structural condition, equipment requirements, etc. Damaged aircraft will be removed from the runway in a minimum time period consistent with the following:

1.1.1.9.1. Personnel safety.

1.1.1.9.2. Prevention of unnecessary secondary damage.

1.1.1.9.3. Preservation of evidence for accident and safety investigation.

1.1.1.10. On-Base and Off-Base Accident Procedures:

1.1.1.10.1. All on-base and off-base accidents will be coordinated in accordance with JBA IEMP 10- 2.

1.1.1.10.2. Under no circumstances will personnel or any equipment be dispatched off base if it jeopardizes the mission of the on-base recovery operation. To dispatch, approval must be granted from the 89 AW/CC and owning organization.

1.1.1.11. In the event an aerospace vehicle is damaged or destroyed, the nearest base with the necessary repair or reclamation capability takes possession. The time of possession change is the time of landing or crash.

1.1.1.11.1. Possession does not change if the parent organization does the repair, reclamation, or termination.

1.1.1.11.2. AVDO (Aerospace Vehicle Distribution Officer) must initiate the proper station location code and possession purpose identifier changes. (Ref DAFI 21-103 section 2.12.3.).

1.2. Terms Defined IAW TO 00-80C-1.

1.2.1. Crashed Aircraft: An aircraft unable to return to designated or alternate field or missed landing resulting in major or total destruction of the aircraft.

1.2.2. Damaged Aircraft: An aircraft that cannot be moved under its own power or supported by its undercarriage without sustaining secondary damage.

1.2.3. Disabled Aircraft: An aircraft that cannot or should not be moved under its own power but can be towed using its own undercarriage.

Chapter 2

ORGANIZATIONAL RESPONSIBILITIES

2.1. IAW DAFI 21-101 AMC Sup 1. IAW DAFI 21-101 AMC Sup 1, the installation wing commander will implement policy, plans and agreements to ensure compliance with established recovery programs.

2.2. 316 WG/CC:

2.2.1. As installation commander, the 316 WG/CC has delegated the JBA CDDAR Program Management to the 89th Airlift Wing (89 AW).

2.2.2. Will ensure that DAFI 10-2501, Air Force Emergency Management Program Planning and Operation, is adhered to for all emergency actions.

2.3. 89 AW/CC.

2.3.1. The 89 AW/CC will oversee implementation of JBA CDDAR policy, plans and agreements to ensure successful execution of the JBA CDDAR program.

2.4. 89 MXG/CC.

2.4.1. Establish and maintain CDDAR capability.

2.4.2. Ensures sufficient equipment is available.

2.4.3. Approve waivers for CDDAR team member training.

2.4.4. Direct recovery/removal operations once IC releases the aircraft. **Note:** PAG/CC is primary authority for recovery/removal operations of PAG aircraft.

2.4.5. Ensure a suitable CDDAR program is established by the Executive Airlift aircraft maintenance contractor and the program executed IAW all applicable USAF, DAFIs, TOs, and the contract PWS.

2.5. 89 MXG Executive Airlift Maintenance Contractor.

2.5.1. Maintain overall responsibility for recovery of host/tenant/transient crashed/damaged/disabled aircraft, on base and off base IAW TO 00-80C-1 & JBA IEMP 10-2.

2.5.1.1. Tenant units will provide their CDDAR team to augment EA Contractor CDDAR team and function as a subject matter expert.

2.5.2. Maintain capability to provide and support recovery operations for all base assigned aircraft, to include tenant unit aircraft.

2.5.3. Manage government furnished weatherproof storage for all recovery equipment.

2.5.4. Perform off-station crash, damaged, or disabled aircraft recovery (CDDAR) for aircraft within a 50 nautical mile radius of JBA-NAFW. Contractor shall respond to the on-scene commander if required.

2.5.5. Lease or purchase specialized equipment and vehicles to support CDDAR effort. This will only be done with the approval of the Functional Area Commander and the Contracting Officer.

2.5.6. On-shift CDDAR team will have qualified members to access the Controlled Movement Area for crossing of an active runway and or taxiway when required to reach recovery site.

2.5.7. Ensure a coordinated response to CDDAR situations by aircraft maintenance personnel, fire protection, and other essential base agencies.

2.5.8. Ensure personnel are adequately trained to prevent injury to personnel, further damage to aircraft, equipment, or other resources.

2.5.9. Review pertinent technical data and previous findings to assess limits of internal unit capability and coordinate with tenants or owning units of mishap aircraft for resources beyond what is possessed.

2.5.10. Appoint a CDDAR Program Manager.

2.5.11. Designate CDDAR Team Chiefs.

2.6. 89 MXG Aircraft Maintenance Contractor CDDAR Program Manager.

2.6.1. Develop initial and annual CDDAR training plan and establish local training program.

2.6.2. Immediately brief the 89 MXG/CC of equipment shortages or serviceability that precludes effective CDDAR support.

2.6.3. Brief 89 MXG/CC on status of CDDAR equipment quarterly, or whenever status changes.

2.6.4. The CDDAR Program Manager serves as the single entry point of contact for external agencies queries regarding CDDAR program coordination.

2.6.5. CDDAR Program Manager will ensure that, Team Chiefs and CDDAR team members remain trained and proficient on CDDAR duties. Training should be documented and tracked by Integrated Maintenance Data System (IMDS).

2.6.6. Manage and direct CDDAR duties IAW DAFI 21-101, DAFI 21- 101_ 89AW Sup, and TO 00-80C-1 relating to CDDAR duties.

2.6.7. Select personnel who will be trained in basic CDDAR operations and function as the core group of personnel from CDDAR response efforts for JBA assigned aircraft.

2.6.8. Select personnel to serve as CDDAR augmentees, as required.

2.6.9. Ensure all personnel are cognizant of hazards associated with aircraft (i.e. composite materials, hydrazine systems, etc.) outlined in TO 00-80C-1. Ensure inspections are performed IAW applicable publications to ensure serviceability of CDDAR equipment. All tools and equipment will be tracked in TCMax. All inspections will be tracked using IMDS and documented on AFTO Form 244.

2.6.10. Ensure team members are assigned and trained to support CDDAR operations. This includes:

2.6.10.1. Basic equipment operation.

2.6.10.2. CDDAR hazardous material and unique characteristic familiarization training for JBA-assigned aircraft (i.e., F-16 hydrazine systems, C-130 ballast depleted uranium, aircraft composite materials, etc.) and for transient aircraft that routinely transit JBA. This training will be documented.

2.6.10.3. Availability and proper use of PPE as determined by the technical data, and BE recommendations during hazardous material response per **paragraph 2.16**.

2.6.11. Ensure special qualifications for personnel are identified and documented. Identify individual team member qualifications for specific equipment operations (i.e., towing, jacking, support equipment, special purpose vehicle).

2.6.12. Maintain a list of all CDDAR tools and equipment.

2.6.13. Ensure tools and support equipment are adequate for recovery, (i.e. bags, consoles, slings, manifolds, PPE/respirators, tow bars, dunnage and shoring (if any), etc.), are serviceable, and available. Ensure the tools and support equipment are inspected at a minimum of every 120 days for serviceability IAW DAFI 21-101_89 AW SUP.

2.6.14. Plan annual JBA CDDAR training events in conjunction with JBA MARE exercises. Coordinate two annual training events to coincide with other similar activities, such as MAREs, to ensure seamless interface with the JBA Incident Response program, ensuring exercise of the CDDAR recovery process prior to ENDEX. Coordinate training with the 89 MXG/CC, 89 AW/SE, 89 AW/XP, and Secret Service offices as well as all CDDAR supported tenant unit maintenance activities.

2.6.15. Coordinate with 89 MXG Aircraft Maintenance Contract QA Weight & Balance manager or owning organization whenever aircraft weight and center of gravity (CG) conditions are unknown.

2.6.16. IAW TO 00-80C-1, coordinate with 316th Civil Engineer Squadron Readiness and Emergency Management (EM) Flight, the EM Working Group and other units as applicable to develop, modify, and implement plans, checklists, instructions and agreements concerning CDDAR operations.

2.6.17. Develop the Aircraft Recovery Plan (ARP) IAW TO 00-80C-1.

2.6.18. Recommend Aircraft Recovery Strategy (ARS) to the IC during CDDAR events.

2.6.19. Assist in the development of a mishap site clean-up plan.

2.6.20. Review all Air Force, MAJCOM and base instructions, functional checklists, base support agreements and base disaster response plans related or pertaining to the local CDDAR program, at least annually.

2.6.21. Identify deficiencies in CDDAR plans, checklists, instructions, and agreements. Address these deficiencies, using established change processes.

2.6.22. Ensure local CDDAR procedures, when developed or changed, are coordinated with the Fire Emergency Services, Safety, Civil Engineer, EM, Explosive Ordnance Disposal, Security Forces, BE, Airfield Manager, and other on- and off-base agencies as required.

2.6.23. Will develop procedures to recall CDDAR team members during non-duty hours.

2.6.24. Identify CDDAR equipment and material that may not be available through base resources; this may include, but is not limited to, heavy equipment, dollies, jacks, and tow vehicles. Develop and maintain procedures to obtain equipment through 89 MXG, 316 CES, 316 CONS, lateral or contract sources if not organically possessed, to include long-term

procuring agreements when required. The CDDAR Program Manager is responsible for identifying and updating all such requirements bi-annually.

2.6.25. Maintain CDDAR trailer and equipment.

2.6.26. Develop and maintain initial CDDAR aircraft recovery checklist, and aircraft removal checklist to protect personnel, aircraft, and equipment using template in TO 00-80C-1 (see **Attachment 2** for CDDAR Aircraft Recovery Checklist).

2.6.27. Establish and maintain an updated continuity book for the CDDAR program IAW TO 00-80C-1.

2.6.28. Maintain a list of vehicles and recovery support equipment identified by JBA units with a flying mission. This list should be provided by JBA units with a flying mission to ensure 24-hour availability.

2.7. 89 MXG Aircraft Maintenance Contractor CDDAR Team Chief.

2.7.1. Be ready to respond to a mishap with the CDDAR team and/or execute CDDAR operations, with little to no notice.

2.7.1.1. The Production Supervisor recalls CDDAR Team Chief when notified by the 89 MXG/MOC. The CDDAR Team Chief will in turn contact other CDDAR members and supporting agencies needed to execute the recovery operation.

2.7.2. Within 60 minutes from notification of an aircraft mishap, utilize CDDAR team recall roster to notify CDDAR team members of potential recovery. Upon notification from IC or ROC (via the MOC) to start recovery, initiate the recall of CDDAR personnel and prepare tools and equipment.

2.7.2.1. The CDDAR Team Chief ensures crash recovery trailer contents are secure for transport, delivers crash trailers to mishap site and maintains tool accountability throughout the mishap. The CDDAR Team Chief ensures delivery of sub-located equipment. CDDAR Team Chief coordinates with base agencies to acquire vehicles and expedite the delivery of tools and equipment not stored in the crash trailers (see paragraph **2.17** and **4.2.1** for 316 CES support requirement).

2.7.3. Be familiar with CDDAR, EM Plans, checklists, instructions, and agreements and understand how they impact the CDDAR team and aircraft recoveries.

2.8. 316 WG Command Post.

2.8.1. Implement the appropriate check lists and notify with the following agencies for 89 MXG Aircraft Maintenance Contractor CDDAR support:

2.8.1.1. 89 MXG Maintenance Operations Center (MOC).

2.8.1.2. 316 CES Readiness and EM Flight (316 CES/CEX).

2.8.1.3. 89th Airlift Wing, Wing Safety (89 AW/SE).

2.8.1.4. 316 MDG.

2.8.1.5. 316 MDG Bioenvironmental Flight (316 OMRS/SGXB).

2.8.1.6. 316th Security Forces Base Defense Operations Center (316 SFG/BDOC).

2.8.1.7. Airfield Management (89 OSS/OSAA).

2.8.1.8. 316th Logistics Readiness Squadron Vehicle Operations (316 LRS).

2.8.1.9. 316th Contracting Squadron (316 CONS).

2.8.1.10. 316th Force Support Squadron (316 FSS).

2.8.1.11. Explosive Ordnance Disposal Flight (316 CES/CED).

2.8.1.12. JBA Fire Emergency Services (316 CES/CEF).

2.8.1.13. Additional on-base or off-base agencies as required.

2.8.1.14. Coordinate with the Emergency Operations Center (EOC), if directed.

2.8.2. In the event of a crashed, damaged, or disabled aircraft, JBA IEMP 10-2 Aircraft Mishap Response will be initiated by the JBA EOC and 316 WG CAT. Once directed, 89 MXG/MOC will initiate Major Aircraft Accident/Incident Check Sheet along with the 316 WG CP and then notify Production Supervisor on duty to assemble the CDDAR Team.

2.9. 89 MXG Maintenance Operations Center (MOC) Senior Controller.

2.9.1. Initiate applicable Quick Reaction Checklist.

2.9.2. Clear one radio (i.e., CDDAR net) for exclusive use by the recovery team to expedite the recovery operation. Monitor radio communication functionality in case additional personnel and/or equipment are required.

2.9.3. Coordinate the needs of the CDDAR Team via radio after they have been activated. All requirements will be coordinated with the IC or Fire Chief while at the mishap site.

2.9.4. In the event of an aircraft being recovered requires the movement of cargo by way of normal or alternate means, contact the 89 APS duty officer for offload.

2.9.5. Maintain a copy of the current appointment letter of the CDDAR Program Manager and CDDAR Team Chief.

2.10. 89 OSS/OSAA (Airfield Management).

2.10.1. Relay all information received from Primary Crash Alarm System (PCAS) via the Secondary Crash Net (SCN). If notified of an emergency other than by the PCAS, the SCN will be activated and will notify the Air Traffic Control Tower (ATCT).

2.10.2. Initiate applicable Quick Reaction Checklists (QRCs).

2.10.3. Respond to all in-flight emergencies and ground emergencies impacting airfield operations except those involving hydrazine. Initiate completion of the hydrazine response QRC if hydrazine is involved.

2.10.4. If notified the aircraft will engage the barrier arresting system, notify Barrier Maintenance.

2.10.5. Determine and inform ATCT of airfield condition and take action to close the runway or affected taxiways.

2.10.6. Make a visual inspection of the runway and affected taxiways for airfield damage or foreign objects. Request Civil Engineer assistance as necessary.

2.10.7. Ensure the runway is clear of all vehicles, equipment, and personnel. Advise ATCT and IC when runway operations can be resumed.

2.10.8. Notify ATCT to divert to alternate runway as necessary.

2.11. 89 AW/SE.

2.11.1. Monitor, assess, and advise on response to aircraft emergencies.

2.11.2. Initiate mishap investigation and reporting as required by DAFI 91-204, *Safety Investigations and Reports,* and form an Interim Safety Board with the owning unit as well as transient unit.

2.11.3. Give guidance for Preservation of Evidence for the Interim Safety Board (ISB) or Safety Investigation Board (SIB).

2.12. 89 AW/XP.

2.12.1. Develop an Incident Action Plan (IAP) for the IC to use during CDDAR operations. Information regarding to the IAP can be found in AFMAN 10-2502 and TO 00-80C-1.

2.12.2. Provide 89 MXG Aircraft Maintenance Contractor with all Support Agreements, Mutual Aid Agreements, MOA, or MOU agreements which include CDDAR support services and any changed agreements thereafter.

2.13. 89 AW/IGI.

2.13.1. Develop a plan for conducting periodic tabletop exercises to discuss possible responses to a variety of scenarios, assess personnel capabilities, exercise checklists, validity of phone numbers, etc. These exercises will mirror other installation major accident response exercises.

2.13.2. Incorporate a CDDAR scenario into one of the two JBA Major Accident Response Exercise per year.

2.14. 89 APS:

2.14.1. Will provide detailed cargo manifest with complete descriptions of any explosives or hazardous materials to IC or others as directed. This includes sourcing manifests for transient inbound crashed, damaged, or disabled aircraft from the owning unit. Provide necessary equipment needed, determined by the CDDAR Team Chief (i.e., forklift, K loader, stair truck, etc.).

2.15. 316 MDG/CC.

2.15.1. Ensure ambulance service responds to a position as directed by the IC with an ambulance and personnel necessary to provide emergency medical care.

2.15.2. Respond with an in-flight emergency or mishap medical team to include a flight surgeon who will dispatch to the scene. The medical team will also respond for incidences involving aircrew or passenger physiological events. Once cleared by the IC, medical responders will ensure the affected personnel are examined and cared for. Medics shall coordinate with 89 AW/SE to ensure medical evaluation covers evidence needed for mishap classifications.

2.16. 316 MDG Bioenvironmental Flight (BE).

2.16.1. Assess health risks, identify occupational, environmental, and radiological hazards, determine training and PPE (for government civilians and active-duty members) requirements, and communicate risks to IC and appropriate individuals at or near the mishap scene. The BE will also make provisions to recall a representative for non-duty hours.

2.16.2. Evaluate the scene for potential health hazards and will provide assessments to the IC.

2.16.3. Maintain surveillance of site conditions and provide updates to the IC and CDDAR Team Chief.

2.16.4. Work with the IC, 89 MXG Aircraft Maintenance CDDAR Team Chief and 316 SFG in determining the peripheral area. The peripheral area should be more than 25 feet away from damaged composite parts, depending on environmental conditions. (Ref TO 00-105E-9, Chapter 3).

2.16.5. Monitor the mishap site for hazardous material exposures and advise the IC.

2.16.6. Brief IC and response personnel on all potential hazards. Specify proper PPE (including respiratory protection) as required based on health risk assessment to appropriate individuals. BE will ensure compliance with 29 CFR 1910.134, Respiratory Protection, as individuals are identified.

2.16.7. Monitor environmental conditions at site during recovery and advise CDDAR team chief of any recommended changes to PPE.

2.16.8. Complete Occupational and Environmental Health (OEH) risk assessments, provide occupational health related protective recommendations and communicate risk to appropriate individuals IAW DAFI 48-145.

2.16.9. Determine the need for monitoring personnel, Contamination Control Station (CCS) procedures, and contamination control requirements with EM Flight as required.

2.17. 316 CES.

2.17.1. Provide manpower and equipment necessary to support the recovery mission as directed by the IC and 89 MXG Aircraft Maintenance CDDAR Team Chief. Assist in providing access to crash site and assist in site setup IAW JBA IEMP 10-2. Make provisions to recall a representative for non-duty hours.

2.17.2. Coordinate delivery of heavy equipment, machinery (i.e., cranes, bull dozers, dollies, jacks, tow vehicle, etc.) and qualified equipment operators as requested by the CDDAR Team Chief, IC or Recovery Operations Chief (ROC).

2.17.3. Procure and deliver necessary supplies needed for the recovery and removal operation (i.e., dunnage, plywood, planking, tools and support equipment). CDDAR Team Chief or ROC are responsible for identifying such requirements in coordination with 316 CES representatives.

2.17.4. Complete a grid survey of the area and identify the location of aircraft parts and remains when directed by the IC or the SIB president.

2.17.5. Direct activities during surface preparation, road building, etc., prior to and during recovery operations.

2.17.6. Maintain a listing of activity-owned equipment required to facilitate pre-recovery planning.

2.17.7. 316 CES/CED:

2.17.7.1. Respond to airfield emergencies to render safe ordnance and aircraft during inflight and ground emergencies or crash situations per AFMAN 32-3001, Explosive Ordnance Disposal (EOD) Program.

2.17.7.2. Assign EOD Team Leader who will direct the use and performance of EOD TTPS's by EOD team members.

2.17.7.3. Whenever system experts are available, EOD personnel should consult, coordinate, and work together with these specialists while ensuring their safety.

2.18. 316 CONS.

2.18.1. When directed by the IC, procure needed supplies and coordinate with the IC and CDDAR Team Chief for availability and delivery of all emergency requests. Supplies may include but are not limited to relief facilities (portable toilets), canopies/tents, mosquito nets, and PPE. Make provisions to recall a representative for non-duty hours.

2.18.2. Develop long-term agreements to obtain access to equipment and material that may not be available through base resources. Such agreements may include but are not limited to obtaining cranes, heavy equipment, dollies, jacks, and tow vehicles.

2.19. 316 LRS.

2.19.1. Maintain process to recall a representative during non-duty hours.

2.19.2. Provide on-scene fuel servicing of recovery support equipment (i.e., AGE and heavy equipment) and provide fuel analysis of aircraft fuel IAW TO 42B-1-1 para 4.10 and AFI 23-201 para., 7.20.

2.19.3. Provide vehicle maintenance support for government-owned vehicles and vehicular equipment during fuel spill clean-up or CDDAR operations when requested by the IC.

2.19.4. Authorize special equipment account for containment and clean up equipment required by JBA IEMP 10-2.

2.19.5. Provide defuel vehicle when requested.

2.20. 316 FSS.

2.20.1. Maintain process to recall a representative during non-duty hours.

2.20.2. Facilitate coordination of lodging, meals, and authorized support services to appropriate personnel identified by the IC.

2.21. 316 CPTS.

2.21.1. Maintain process to recall a representative during non-duty hours.

2.21.2. Establish a fund site to procure needed equipment and supplies necessary to execute CDDAR recovery operations.

2.22. 316 SFG.

2.22.1. At the request of the IC, respond to emergencies with sufficient personnel and vehicles to provide initial security for a crash site or damaged aircraft and to maintain security of classified equipment. SF personnel will respond with appropriate PPE, lethal and non-lethal capabilities for self-protection and the protection of others.

2.22.2. Follow directions given by IC to limit access and secure aircraft or crash site.

2.22.3. Establish a cordon area and ECP (entry and exit control point) in conjunction with the Fire Chief, BE and IC. The cordon size may expand as the situation warrants (Ref TO 00-105E- 9, Chapter 3).

2.22.4. When required, request the cooperation of police and civil defense authorities to handle extraordinary circumstances that may develop with an off-installation aircraft mishap.

2.22.5. Adhere to Posse Comitatus Act during off-base mishap responses if a National Defense Area has not been established.

2.23. 316 WG/JA.

2.23.1. Maintain process to recall a representative during non-duty hours.

2.23.2. Provide legal advice to the IC. Legal support shall include, but is not limited to, the following actions:

2.23.2.1. Advise the IC on necessary steps to preserve accident or safety investigation evidence.

2.23.2.2. Determine legal jurisdiction and recommend to the IC the need to establish a National Defense Area (NDA).

2.23.3. Determine affected property owners and obtain right of entry onto private land as necessary.

2.23.4. Assess the situation to determine potential third-party claims and assess the need to establish a temporary claims office. Establish a temporary claims office as circumstances dictate.

2.23.4.1. Photograph and inspect damage to private property as soon as practical with permission and coordination of the Safety Investigation Board.

2.23.5. Provide immediate legal services to personnel involved in accident/incident.

2.23.6. Review and approve MAAs before finalization and implementation.

2.23.7. Be prepared to represent the government for an off-base aircraft mishap.

2.24. JBA Units with Flying Missions.

2.24.1. Tenant unit CDDAR team members shall be under the tactical control (TACON) of the 89 MXG Aircraft Maintenance CDDAR Team Chief during CDDAR recovery operations.

2.24.2. Coordinate with 89 MXG Aircraft Maintenance Contractor CDDAR team and participate in exercises, training, and equipment inventories.

2.24.3. Review support agreements to help assess limits of internal unit capabilities and coordinate with the 89 MXG Aircraft Maintenance Contractor for resources over and above those organically possessed.

2.24.4. Ensure personnel are prepared to assist and provide expertise in CDDAR situations. As a minimum, ensure personnel are trained on procedures for responding to more common incidents that would require CDDAR, (i.e., blown or flat tires, aircraft departing prepared surfaces, and major fuel spills).

2.24.5. Be prepared to assign a Subject Matter Expert to perform Recovery Operations Chief duties (ROC).

2.24.6. Assist 89 MXG Aircraft Maintenance Contractor recovery operations. Tenant shall provide:

2.24.6.1. Technical Expertise.

2.24.6.2. Technical Data.

2.24.6.3. MDS-unique tools and special equipment.

2.24.6.4. Airframe and system familiarization.

2.24.6.5. Manpower and augmentation as needed, to include necessary support for hazardous material response for all aircraft assigned to or transiting JBA.

2.24.6.6. Qualified CDDAR team members for each tenant unit MDS. At least one tenant unit member shall be CDDAR qualified, and they shall fulfill the Assistant CDDAR Team Chief duties during recovery operations on tenant unit owned aircraft.

2.24.7. Identify vehicles and recovery support equipment in a local directive to ensure 24-hour availability and report capabilities to 89 MXG/CC and CDDAR Program Manager.

2.24.8. Develop procedures to recall CDDAR team members during non-duty hours to provide CDDAR recovery operations assistance.

Chapter 3

CDDAR TEAM QUALIFICATIONS AND SPECIALTY REQUIREMENTS

3.1. General.

3.1.1. All CDDAR team members and tenant unit CDDAR personnel must be qualified in basic CDDAR operations (except augmentees).

3.2. CDDAR Qualifications.

3.2.1. Basic CDDAR qualifications include aircraft towing, tire removal/change, aircraft jacking, aircraft defuel, GOX/LOX de-servicing.

3.2.2. CDDAR qualifications shall be recorded in employee personnel training records and entered into applicable MIS, for all CDDAR team members, 89 MXG Aircraft Maintenance Contractor and tenant units.

3.3. Training Requirements.

3.3.1. All recovery team members must receive initial training comprised of both academic and hands-on training. Personnel used to augment real-world recoveries do not require CDDAR specific training. For the CDDAR Team Chief, training will be provided through an AETC formal training course. The Team Chief will then train all other members IAW TO 00-80C-1 and DAFI 21-101_89 AW SUP.

3.3.2. Aircraft lift training is considered advanced CDDAR operations. If training with owned aircraft, units may position lifting bags and equipment, but will only inflate bags so the bags mate to the aircraft. Extreme caution will be used to prevent damage to the aircraft or equipment. Under no circumstances will units attempt to lift an active aircraft off the ground.

3.3.3. All recovery team members must receive annual training compromised of both academic and hands-on training. Hands-on training includes aircraft lifting exercises using a unit owned aircraft or Ground Instructional Training Aircraft (GITA). Do not use operational aircraft for actual aircraft lifts in a training environment. If units have no available training assets, consider participating with other organizations possessing training assets. If no assets are available or suitable for these exercises, units demonstrate capability by completing all steps but stopping short of actually lifting an operational aircraft. Ensure all training is documented in applicable MIS.

3.3.4. CDDAR team chiefs must complete an actual aircraft lift every 3 years. This requirement can be accomplished via real-world recovery events, the AETC formal training course or lifting a GITA.

3.3.5. The owning unit MXG/CC or equivalent may waive training requirements as circumstances dictate. Waivers must not be used in lieu of training if training is available. Units must make every effort to schedule personnel consistent with this requirement.

3.4. Specialty Requirements.

3.4.1. There are several aircraft recovery tasks (i.e., defueling, towing, jacking, servicing, etc.) which require detailed knowledge of specific aircraft systems, components, and their operational function(s). Given the requirements to recover aircraft as quickly as possible and

without causing secondary damage, the CDDAR team must include personnel with sufficient technical knowledge and crash recovery knowledge to rapidly execute the aircraft recovery. CDDAR Team Chiefs will ensure CDDAR team members meet the minimum Air Force maintenance specialty requirements specified by TO 00-80C-1. The actual number of CDDAR team members used to execute an aircraft recovery operation will depend on the situation and circumstances of the crashed, damaged, or disabled aircraft being recovered. CDDAR teams shall be robust enough to accommodate multiple shifts and long duration recovery efforts. CDDAR teams may also include augmentee personnel from specialties not listed in TO 00-80C-1.

Chapter 4

CDDAR VEHICLE AND SUPPORT EQUIPMENT REQUIREMENTS

4.1. General.

4.1.1. The 89 MXG/CC approves vehicle/equipment CDDAR requirements. CDDAR Program Manager shall identify vehicle/equipment requirements to 89 MXG/CC. Tenant units will identify available CDDAR vehicles and recovery support equipment in a local directive. They will also provide availability of each CDDAR designated vehicle and equipment items (i.e., available 24/7) and contact information to secure CDDAR designated vehicles and equipment. Once published, a copy of this local directive shall be distributed to the 89 MXG CDDAR Program Manager. Units shall have sufficient aircraft lifting equipment to accomplish a complete list of their assigned MDS aircraft. (Exception: PAG is not required to have lifting equipment for VC-25 aircraft). This requirement can be met by any combination of lift bags, aircraft jacks, or aircraft slings. It is not necessary to maintain sufficient lift bags to lift an entire aircraft using only lift bags unless otherwise directed by an aircraft technical order.

4.2. Minimum Vehicle/SE requirements.

4.2.1. 316 CES provides:

4.2.1.1. All Terrain Forklift.

4.2.1.2. Light Carts

4.2.1.3. Dunnage/shoring

4.2.2. 89 MXG provides:

4.2.2.1. Aircraft tow vehicle.

4.2.2.2. Tow bars.

4.2.2.3. Air bags.

4.2.2.4. Slings, belly bands, snatch cables, chains, etc.

4.2.2.5. Aircraft jacks.

4.2.3. Tenant Units provide:

4.2.3.1. MDS specific support equipment.

Chapter 5

JBA DISASTER RESPONSE FORCE STRUCTURE

5.1. General.

5.1.1. In the event the 316 WG/CC has not already activated the JBA Crisis Action Team (CAT) for other response efforts, the 89 AW/CC, as the delegated CDDAR execution agent, may activate the JBA CAT IAW TO 00-80C-1 to manage the overall incident response operation. See AFMAN 10-2502 and TO 00-80C-1 for CAT information flow.

5.2. Crisis Action Team (CAT).

5.2.1. If the JBA CAT is activated solely to support CDDAR operations, 89 AW/CC will serve as the CAT Director. The CAT will be the primary point of contact between responding installation, higher headquarters entities, and all outside agencies.

5.2.2. The CAT, through the Command Post (CP), sends and receives information and requests which are pertinent to emergencies. When notification of a major accident is received, the CP activates the Installation Notification and Warning System (INWS), notifies all base organizations, and notifies off-base agencies supported by the installation if required.

5.2.3. The CAT director will consider activating EOC, contacting affected senior leaders and recalling the Disaster Response Force (DRF) based on the CDDAR scenario at hand.

Chapter 6

CRASH/MISHAP RESPONSE CHRONOLOGICAL PROCEDURES

6.1. First Responders.

6.1.1. 316 WG CP notifies first responders (i.e., Fire Emergency Services, Emergency Medical Services and 316 SFG). During incidents, first responders proceed to the scene and secure the immediate incident area, establish Incident Command System (ICS), provide rescue and firefighting, identify and contain hazards, and provide patient care, triage, medical monitoring, transport, and decontamination procedures. See AFMAN 10-2502 and TO 00-80C-1 for ICS model.

6.1.2. The IC assumes control of the aircraft recovery operations. Individuals will report to the IC before approaching or performing any actions on the aircraft.

6.2. Emergency Responders.

6.2.1. When requested by the IC, the 316 WG CP or JBA EOC (if activated) will notify remaining emergency responders.

6.2.2. Deploy to the accident scene after the first responders to expand command and control and perform support functions as directed by EOC Director or IC.

6.2.3. Consists of additional firefighters, security forces, and emergency medical technicians, as well as EM personnel, EOD personnel, physicians, nurses, medical treatment providers at medical treatment facilities, public health officers, BE, and mortuary affairs personnel.

6.3. Specialized Teams.

6.3.1. When requested by the IC, the 316 WG CP or JBA EOC activates/tasks the 89 MXG Aircraft Maintenance Contractor CDDAR team.

6.3.1.1. When tasked, the 89 MXG Aircraft Maintenance CDDAR Program Manager will deploy essential personnel and equipment to the mishap site. If required, the CDDAR Team Chief will contact tenant unit CDDAR teams for recovery operation assistance.

6.3.1.1.1. During initial response, if/when requested by the IC, the CDDAR Team Chief will respond to the mishap site.

6.3.1.2. The 89 MXG Aircraft Maintenance CDDAR Team Chief will take direction from the IC and the ICS chain of command until the IC has transferred authority to the ROC.

6.3.1.3. The 89 MXG Aircraft Maintenance CDDAR Team Chief directs the recovery team and will use the following radio call signs on the established net (i.e., CDDAR net).

6.3.1.3.1. Recovery Team Chief (Recovery 1).

6.3.1.3.2. Assistant Recovery Team Chief (Recovery 2).

6.3.1.3.3. Recovery Tow Supervisor (Recovery 3).

6.3.1.3.4. Recovery Tow Team Driver (Recovery 4).

6.3.1.4. The 89 MXG Aircraft Maintenance CDDAR Team Chief will evaluate the situation, to include safety, aircraft damage, structural integrity, weight, terrain, and current

weather conditions. Available time for reclamation operations will also be a consideration. Basic reclamation operations will be accomplished according to aircraft specific technical data whenever available.

6.3.1.4.1. Before entering the mishap area for scene evaluation, the CDDAR Team Chief will don proper PPE (i.e., hardhat, eye protection, safety glasses, respirator, etc.).

6.3.2. When requested by the IC, JBA EOC recalls the on-call 89 AW/PA and 316 WG/PA photographers to videotape or photograph the crash site to give authorities an overview of the crash site.

6.3.2.1. Photograph and catalog any aircraft material before removal from crash site. If wreckage is interfering with essential mission activities or poses an extreme hazard, the installation commander in coordination with the ISB or SIB president may choose to remove or move wreckage.

6.3.3. After the SIB President releases the aircraft, the ROC will request 89 MXG Aircraft Maintenance Contractor to collect and ensure safety of the aircraft. The ROC will continue to assume responsibility for the recovery, reclamation, and removal process and will ensure any further actions needed on-scene are completed.

Chapter 7

CDDAR AIRCRAFT RECOVERY/RECLAMATION/REMOVAL

7.1. Procedures.

7.1.1. Before recovery actions, the IC will inform the 89 MXG Aircraft Maintenance CDDAR Team Chief of the following:

7.1.1.1. The required PPE needed at the scene. BE will determine the necessary level of PPE for, at, or near the mishap scene per **para 2.15**.

7.1.1.2. Check the surface under/around aircraft, wind direction/velocity, and location near any other elements that may compromise safety.

7.1.1.3. Verify no hydrazine leak exists. If suspected, only qualified personnel equipped with self-contained breathing apparatus (SCBA) will approach the aircraft to conduct evaluation. Containment and clean-up will be performed by a specialized Hydrazine Response Team.

7.1.1.4. Verify that the aircraft is safe to approach by inspecting the aircraft for stability, prior to the CDDAR Team Chief installing safety devices on the aircraft to the greatest extent possible.

7.1.1.5. Verify there are no hazardous fluid leaks (e.g., fuel, oil, hydraulic fluid, etc.). Notify CDDAR Team Chief if hazardous condition exists.

7.1.1.6. Verify that munitions are safe and no hazardous material/cargo condition exists. Notify CDDAR Team Chief if hazardous condition exists.

7.1.2. The 89 MXG Aircraft Maintenance CDDAR Team Chief coordinates with the IC to ensure the aircraft is safe before any member approaches the aircraft. The IC, as part of this procedure, will have checked to ensure there are no hazardous leaks or hazardous situations such as hot brakes etc. Once the aircraft is deemed safe, and the IC is no longer required, the aircraft will be released to the ROC and CDDAR Team Chief for recovery.

7.1.2.1. CDDAR team members will only take direction from the 89 MXG Aircraft Maintenance CDDAR Team Chief who takes direction from the ROC after the aircraft has been released by the IC for recovery.

7.1.3. The 89 MXG Aircraft Maintenance CDDAR Team Chief distributes specific tasks to CDDAR team members and establishes an assembly point.

7.1.4. The 89 MXG Aircraft Maintenance CDDAR Team Chief will brief the IC and the CDDAR team members, and, if appointed, the ISB and/or SIB president of the reclamation plan, benefits, limitations and anticipated hazards before beginning the recovery operation.

7.1.5. After situation assessment, the 89 MXG Aircraft Maintenance CDDAR Team Chief will conduct a detailed safety briefing with all CDDAR team members. At a minimum, the safety briefing will include:

7.1.5.1. Aircraft condition.

7.1.5.2. Hazard awareness such as presence of running engines, arresting gear, barrier cables, sharp edges, flammable fluids, toxic fluids, noxious fluids, high temperatures, composite fibers, blood borne pathogens and hoisting dangers when applicable.

7.1.5.3. Team member duty assignments and responsibilities.

7.1.5.4. Need for continuous situational awareness and buddy-care.

7.1.5.5. Emphasize safety over speed.

7.1.5.6. Expected actions, timeline, and end goal.

7.1.5.7. Wind speed/direction and awareness to changes.

7.1.5.8. Proper display of line badges within a restricted area when necessary.

7.1.5.9. PPE required to enter the recovery area.

7.1.6. After accomplishment of the safety briefing, the recovery team will evaluate the aircraft and create an aircraft recovery plan. The 89 MXG Aircraft Maintenance CDDAR Team Chief will advise the ROC and request concurrence.

7.1.7. 316 SFG will maintain an appropriate cordon around the mishap site during recovery operations. 316 SFG will receive personnel entry approval from the IC or ROC. If 316 SFG are not able to supply a physical cordon, the ROC will direct all non-essential personnel to remain clear of the aircraft.

7.1.8. The 89 MXG Aircraft Maintenance CDDAR Team Chief will accomplish recovery/reclamation/removal procedures IAW TO 00-80C-1 and any applicable aircraft technical data available.

7.1.9. Throughout the recovery, removal, or reclamation operation, if explosive components or human remains are discovered, the reclamation will immediately cease.

7.1.9.1. EOD will render safe and remove all explosive or ejection components (i.e., squibs, seat sequencers, catapults, or initiators).

7.1.9.2. Notify ROC to request BE for suspected or known Chemical, Biological, Radiological, and Nuclear (CBRN) hazards such as depleted uranium, hazardous cargo, advanced composite materials, and hydrazine. (Note: 113th Wing, Air National Guard, has ultimate authority on all hydrazine spills on JBA.)

7.1.9.3. Notify ROC to request Mortuary Affairs for recovery of human remains if found during the recovery and reclamation process.

Chapter 8

RESPONSE TO COMPOSITE MATERIALS

8.1. Initial Response to Composite Materials.

8.1.1. IC will direct the evacuation of all unprotected personnel away from the crash site. Only the Fire Department with appropriate SCBA fire protection equipment is authorized near the mishap site.

8.1.2. IC will inform any potentially affected population to remain inside, shut all windows and doors and turn off air handling and conditioning systems and await further guidance.

8.1.3. BE conducts an initial survey and performs a health risk assessment to determine if air monitoring or sampling is warranted for CBRN materials. Air monitoring will continue through the recovery phases to evaluate airborne hazard potential if release is suspected.

8.1.4. Cordon site and establish single entry control point. Cordon determined by IC and the BE; 316 SFG executes.

8.1.5. Fire Emergency Services assesses the incident, consults with aircraft and BE representatives, and provides recommendations for recovery, storage, and disposal of composite material or other hazardous or contaminated materials.

8.1.6. IC coordinates with Air Traffic Control to limit and/or restrict all flights.

8.1.7. Responders and support personnel on scene will not eat, drink, chew gum, or use any tobacco products of any kind when operating within or near the cordoned area. The only exception is the provision of drinking water outside the cordon to compensate for heat stress. Personnel will be advised to wash their hands, face, and rinse off water containers prior to consumption.

8.1.8. All intended actions regarding composite materials will be briefed to the IC.

8.1.9. The 113 WG will provide the 89 MXG Aircraft Maintenance Contractor CDDAR team with members trained/qualified to handle composite materials in the event of composite material hazards.

8.2. Composite Material Handling.

8.2.1. Purpose: This section contains basic response procedures for aircraft mishaps involving advanced aerospace materials to minimize the associated environmental, safety, and health hazards.

8.2.2. These guidelines are generic in nature and provide basic overview of procedures.

8.2.3. Several aircraft with composite materials frequent JBA, such as, C-17, F-15, F-16, and modern civilian transport aircraft. Use the specific aircraft technical order or consult the TO 00-105E-9 for composite hazards.

8.2.4. BE will inform the IC, 89 MXG Aircraft Maintenance CDDAR Team Chief, and recovery members of specific hazards present and will advise on specific types of personal and respiratory protective equipment required for personnel entry into the scene. The 89 MXG Aircraft Maintenance Contractor CDDAR Team provides clean up and containment of advanced composite materials. These materials, when burned, are capable of emitting highly

toxic vapor and airborne particles. Material emissions cause severe damage to the respiratory system. CDDAR team members will wear specialized protective clothing for this function.

8.2.5. PPE Required:

8.2.5.1. Burning or smoldering composites.

8.2.5.1.1. SCBA.

8.2.5.1.2. Full Protective Clothing (type of aircraft will determine which ensemble is required).

8.2.5.2. Broken, burnt or splintered composites.

8.2.5.2.1. NIOSH-approved full-face respirator with N-100, R-100 or P-100 rated particulate cartridges.

8.2.5.2.2. Hooded, coated Tyvek suit with booties.

8.2.5.2.3. Nitrile rubber gloves (inner).

8.2.5.2.4. Leather work gloves (outer).

8.2.5.2.5. Steel Toe Boots.

8.2.5.3. Peripheral area composite exposure.

8.2.5.3.1. Long sleeve work uniform.

8.2.5.3.2. Safety glasses with side shields or goggles.

8.3. Containment of Composite Materials.

8.3.1. The Composite Material Containment Team secures burned and mobile composite fragments, loose ash and particulate with plastic, firefighting agent, fixing material or a tent. The common fixing agent is an acrylic floor wax and water mixture. Consult with specific aircraft authorities and investigators before applying. Critical safety concerns and investigation concerns may override this use.

TODD E. RANDOLPH, Colonel, USAF
Commander, 316TH Wing

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFPD 21-1, *Maintenance of Military Materiel*, 01 August 2018

DAFI 91-204, *Safety Investigations and Reports*, 10 March 2021

DAFI 21-101 AMC SUP, *Aircraft and Equipment Maintenance Management*, 03 February 2022
DAFI 21-103, *Equipment Inventory, Status, and Utilization Reporting*, 30 April 2020

DAFMAN 32-3001, *Explosive Ordnance Disposal Program*, 22 April 2022

DAFI 91-202, *The US Air Force Mishap Prevention Program*, 12 March 2020

DAFMAN 91-203, *Occupational Safety, Fire, and Health Standards*, 25 March 2022

JBA OPLAN 91-204, *Mishap Response Plan*, 15 April 2012

JBA IEMP 10-2, *Installation Emergency Management Plan*, 1 October 2021

TO 00-105E-9, *Aircraft Emergency Rescue Information*, 16 April 2018

TO 00-80C-1, *Crashed, Damaged, Disabled Aircraft Recovery Manual*, 17 November 2022

TO 00-20-1, *Aerospace Equipment Inspection, Documentation, Policies, and Procedures*, 26 September 2022

DAFI 10-2501, *Air Force Emergency Management Program*, 10 March 2020

DAFMAN 10-2502, *Air Force Emergency Management System (AFIMS) Standards and Procedures*, 13 September 2018

DAF 847, *Recommendation for Change of Publication*, 15 April 2022

DAFI 33-322, *Management of Records*, 28 July 2021

DAFI 32-2001, *Fire Emergency (FES) Program*, 28 July 2022

DAFI 25-201, *Intra-Service, Intra-Agency, and Inter-Agency Support Agreements Procedures*, 18 October 2013

AFTO 244, *Industrial/Support Equipment Record*, 21 April 2017

29 CFR 1910.134, *Personal Protective Equipment - Respiratory Protection*

DAFI 48-145, *Occupational and Environmental Health Program*, 22 September 2022

DAFI 23-201, *Fuels Management*, 09 August 2021

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

AGE—Aerospace Ground Equipment

APS—Aerial Port Squadron

AVDO—Aerospace Vehicle Distribution Officer

ATCT—Air Traffic Control Tower

BE—Bioenvironmental Engineering

CONS—Contracting Squadron

CDDAR—Crashed Damaged/Disabled Aircraft Recovery

CES—Civil Engineer Squadron

CPTS—Comptroller Squadron

DRF—Disaster Response Force

EOD—Explosive Ordinance Disposal

HQ AMC—Headquarters Air Mobility Command

IC—Incident Commander

IEMP—Installation Emergency Management Plan

LRS—Logistics Readiness Squadron

MAA—Mutual Aid Agreements

MARE—Major Accident Response Exercise

MDG—Medical Group

MDS—Mission Design Series

MOC—Maintenance Operations Center

MXG—Maintenance Group

MXG/CC—Maintenance Group Commander

PAG—Presidential Airlift Group

PCAS—Primary Crash Alarm System

PLS—Presidential Logistics Squadron

PPE—Personal Protective Equipment

QRC—Quick Reaction Checklist

ROC—Recovery Operations Chief

SCN—Secondary Crash Net

SFG—Security Forces Group

SIB—Safety Investigation Board

UCC—Unit Control Center

VIPSAM—Very Important Person Special Air Mission

Attachment 2

CDDAR AIRCRAFT RECOVERY CHECKLIST (POC: TEAM CHIEF)

Figure A2.1. CDDAR Aircraft Recovery Checklist (POC: Team Chief).

Aircraft Tail: _____ Time Notified: _____ Date: _____

A. Initial Information:

1. Initial information on crash/mishap and location, utilizing CDDAR-1 Checklist.
(Give pertinent facts which are known at present time).

2. Known Hazards, utilize CDDAR-Aircraft Preparation Checklist and CDDAR-8 Make Aircraft Safe for Maintenance:

a. Wind directions/velocity:
b. Verify aircraft safe to approach:
c. Area safe:
de. Condition of aircraft:
e. Known Hazards (Hydrazine/Batteries/Oxygen/Hydraulic):
f. Munitions:
g. Hazardous Cargo:
h. BE Requirements:

3. Area Cordoned (316 SFG):

4. Key Personnel:

a. Name of IC:
b. Name of ROC:
c. Name of CDDAR Team Chief:
d. Name of BE Representative:

B. Crash/Mishap Response: When requested by IC, 316 WG CO. or JBA EOC activate/task 89 MXG Aircraft Maintenance Contractor CDDAR Team.

1. CDDAR Team Chief: Sign out Recovery1 Radio, Report to IC/Mishap Site for Aircraft Situational Brief within 1 Hour when requested by IC:

2. CDDAR Team Chief: will take direction from the IC and the ICS chain of command.

3. CDDAR Team Chief: will evaluate situation, to include safety, aircraft damage, structural integrity, weight, terrain, weather conditions. Available time required for reclamation operations.

4. Assemble CDDAR Team (Ensure all Members have line Badge:

a_____	g_____
b_____	h_____
c_____	i_____
d_____	j_____
e_____	k_____
f_____	l_____

5. Preliminary Deployment Actions:

a. Team Chief receive aircraft situational brief from IC/ROC. (**Get radio from IC for direct communications from IC/ROC).**

b. Sign out CDDAR Radios: **CDDAR personnel will coordinate on CDDAR Net Call Sign Recovery 1/2/3/4.**
c. Brief aircraft situation to CDDAR Team and assign initial assignment duties.
d. Give initial Safety Brief (Utilize CDDAR Safety Brief Checklist).
1. Equipment and pre-deployment prep to crash site:
2. Prepare CDDAR Trailer and equipment for transport.
3. Ensure trailer and tool kits are signed out.
4. **BUY BOTTLED WATER** to take with you.
5. Sign out E-TOOL, extra batteries and plug-in chargers
6. All Vehicle and Recovery Equipment, AF Form 1800 signed off. (For over the road deployment coordinate with MOC/Vehicle Maintenance).
e. Notify MOC CDDAR Team assembled. **Time notified:**
f. Standby until IC Releases the aircraft:
g. 89 MXG/ROC will request CDDAR to assume responsibility for the recovery/reclamation/removal process.
C. **Recovery:**
1. **CDDAR Aircraft Recovery/Reclamation Plan:** Before recovery operations, CDDAR Team Chief and CDDAR Manager will coordinate with IC/ROC to evaluate scene for following factors:
2. Location and availability of PPE. Coordinate with BEE or Appointed Rep. for required PPE to be used at the site.
3. Conditions around under/around aircraft, wind direction/velocity and location near any elements that may compromise safety.
4. Verify no Hydrazine leaks, aircraft safe to approach by inspecting for stability and installing safety devices. No hazardous fluid leakage. No Munitions or hazardous cargo exist. Notify IC/ROC if hazardous conditions exist.
5. Designate Emergency Rally point:
6. Give Safety Brief (Utilize CDDAR Safety Brief Checklist) for each phase of recovery.
7. Assign task duties and update status board at each step of recovery process.
8. Utilize Access Control Log: Team members name and time when entering and exiting site.
D. **Aircraft Recovery Plan:**
a. Utilize supporting technical references; Aircraft Maintenance Manual, JBANDREWSI21-105 and T.O. 00-80C-1 checklist CDDAR-1 thru CDDAR-8 to develop Aircraft Recovery Plan. **Ensure FSR concurrence.**
b. Review Aircraft Forms (IF Possible).
c. Inform QA of aircraft condition and configuration for weight and balance/CG computation if unknown.
ATTENTION: Must be paid to properly protecting, recovering, securing and removing classified information and materials and other operationally sensitive information from the mishap site. The IC/ROC shall establish the onsite procedures that are required to accomplish this objective. These procedures should be in place prior to CDDAR Team entering site.
E. Post Recovery:
a. Upon completion of recovery operation: Assign task duties and update status board to clean up recovery site and remove equipment from area. Log in team member's name and time when entering and exiting site.

b. Give Safety Brief (Utilize CDDAR Safety Brief Checklist).
c. Return all recovery equipment to 89 MXG.
d. Clean, inspect and recertify all CDDAR Recovery Equipment prior to storage.

Attachment 3

SAFETY BRIEF CHECKLIST

Figure A3.1. Safety Brief Checklist.

Aircraft condition.
Hazard awareness such as:
Presence of running engines, arresting gear Barrier cables.
Sharp edges.
Flammable/toxic or noxious fume fluids.
High temperatures, remember to **hydrate** and adhere to rest cycles Composite fibers, ensure required PPE.
Blood borne pathogens.
Hoisting dangers when applicable.
Team member's duties and responsibilities.
Need for continuous situation awareness and buddy care, no one should be in the crash site alone.
Emphasize safety over speed.
Expected actions, timeline and end goal.

Required PPE:

Hard hats and work gloves must be worn at all times.
Hearing protection will be used when exposed to high noise levels. Respiratory protection will be used when directed by the BE or Team Chief.
Wear leather gloves when working on site and especially when handling sharp and jagged objects.
Safety toe footwear will be worn at all times in crash site.
Authorized and **approved eye protection** will be required and used when directed by Team Chief.
Fall protection will be worn when required or directed by the Team Chief.
Remove all Jewelry (rings, necklaces, bracelets, earrings, wrist watches). **No Cell Phones** will be used except Government Phones used for the recovery process.
Every team member has responsibility to call **"Knock it Off"** if something does not look right. The recovery will be stopped, and concern addressed before resuming.
-Safety is the number one priority. Be aware of your surroundings and watch for potential hazards.
-No one is allowed under any aircraft being suspended with airbags, unless directed by CDDAR Team Chief.
-In an emergency, evacuate from the site, leave area immediately and report to a pre-designated assembly location.
-CDDAR Team members will take direction only from the CDDAR Team Chief and the CDDAR Team Chief will take direction only from the IC/ROC.